FossilJet: AI-Driven Farm Optimization

Purpose

The presentation outlines a technology-driven agricultural solution that leverages IoT, AI, and big data to improve farm productivity, conserve resources, and support small farmers in India and beyond.

Key Takeaways

- The system delivers actionable intelligence to farmers via mobile and WhatsApp, accessible even to non-literate users through audio.
- Over **14,000 IoT devices** have been deployed across **200,000 acres** in India and six countries, collecting **10 billion data points** over 5-7 years.
- The technology enables automated **irrigation and fertigation** without labor, resulting in significant environmental savings: **83 billion liters of water**, **210,000 kg of avoided pesticides**, and **56,000 metric tons of GHG emissions** reduced.
- A business model focused on **hardware sales (CAPEX)** with minimal subscription (OPEX) aligns better with farmer preferences.
- Core algorithms predict crop stage, irrigation needs, and manage **200+ disease and pest models**, with climate risk models applicable globally.

Technology and Impact

The solution, FossilJet and FossilJet Pro, uses real-time data from IoT sensors measuring soil moisture, canopy temperature, and leaf wetness. This data powers AI models that guide precise irrigation and fertilizer application. The technology is crop-agnostic and based on universal biological principles, requiring minimal adaptation across regions.

Yield improvements range from **15% to 300%**, depending on crop and baseline practices, with recovery of technology costs within a year for horticulture crops. Water savings are **at least 30% per farm**, and **15% fertilizer reduction** is estimated, though not yet audited.

Business Model

The company sells hardware outright, having shifted from a subscription model due to low farmer adoption of recurring payments (OPEX). A small subscription component remains over the hardware's lifecycle. The focus on horticulture is driven by higher per-acre value, which justifies the investment.

Data and Future Applications

The vast dataset-larger than India's official IMD weather network-has potential for commercial use in supply chain forecasting, carbon and water credits, and agricultural planning. For instance, real-time crop health data helps predict market supply, guiding buyers and input suppliers. The team is exploring these monetization paths but is not currently commercializing the data.

Global Scalability

Despite regional crop variations, the scientific fundamentals are universal. Only **about 15% of the work** requires localization, primarily for pest and disease resistance. R&D over four to five years ensured globally scalable models from the outset.

Action Items / Next Steps

- Scale deployment from **13,000 to 20,000 farms** by the end of the financial year.
- Explore commercial applications of aggregated farm, climate, and crop data for carbon and water credits.
- Expand into additional markets, leveraging existing models with minimal customization.
- Continue refining algorithms for disease, pest, and climate risk management.